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A New Approach to the Economic Analysis of
Nonstationary Time Series and the Business Cycle<https://www.jstor.org/stable/1912559>

p. 357-384

James D. Hamilton

This paper proposes a very tractable approach to modeling changes in regime. The parameters of an autoregression are viewed as the outcome of a discrete-state Markov process. For example, the mean growth rate of a nonstationary series may be subject to occasional, discrete shifts. The econometrician is presumed not to observe these shifts directly, but instead must draw probabilistic inference about whether and when they may have occurred based on the observed behavior of the series. The paper presents an algorithm for drawing such estimation of population parameters by the method of maximum likelihood and provides the foundation for forecasting future values of the series. An empirical application of this technique to postwar U.S. real GNP suggests that the periodic shift from a positive growth rate to a negative growth rate is a recurrent feature of the U.S. business cycle, and indeed could be used as an objective criterion for defining and measuring economic recessions. The estimated parameter values suggest that a typical economic recession is associated with a 3% permanent drop in the level of GNP.

[Cite This Paper](#)

Journal News

December 11, 2025

In Memoriam:
Kate Ho

We are deeply saddened
by the passing of Kate
Ho, the John L. Weinberg
Professor of Economics
and Business Policy at

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